Quantum-integrability scheme condition for the canonical good Boussinesq equation

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Abstract

(c) The elastic counterterm is tangent to an explicitly conserved family. For $H_s = \int[\pi^2/2 + su^2/2 - u^3 + u_x^2/2]\mathrm{d}x$, the first higher charge can be chosen as $Q_s = \int[\pi_x^2 + u_{xx}^2 - 3u\pi^2 + 3u^4 - 9uu_x^2 + s(u_x^2 - u^3)]\mathrm{d}x$. Its canonical bracket with H_s vanishes for every s. Consequently $s = 1 + 2\hbar\lambda$ admits the correction $2\lambda\int(u_x^2 - u^3)\mathrm{d}x$ for every λ. The order-$\hbar$ local integrability condition cannot determine this finite coupling.

(c) Free-good-vacuum Wick ordering alone fails to conserve the charge. In the stated contraction-cutoff prescription its defect is removed by local composite-charge mixing $18\hbar A_\Lambda \int :u^2 : \mathrm{d}x$, with $A_\Lambda = (\sqrt{1+\Lambda^2}-1)/(2\pi)$, independently of λ. This is a formal local Ward identity, not a construction of a convergent continuum quantum operator. The absence of a λ solvability condition also follows without assuming that such an operator exists.

(c) Static modulus matching independently gives $\lambda = 9/(4\pi)$; zero mean strain and zero reference pressure alone do not. Conditional one-body and two-body coefficients become λ independent after consistent sound-unit and action normalization. **(d)** Use static matching provisionally for the requested one-loop light–light phase, pending the recorded PI decision. The absolute phase is still to be calculated.

1 Scope, conventions and verification

Labels mean: (a) established in literature, (b) literature heuristic, (c) project derivation/check, (d) proposal or unestablished identification. Every displayed derived identity is covered by the script named in its subsection. Scripts are in `phase2/scripts/`; numerical tests use 60 or 70 decimal digits. Finite samples do not prove operator convergence. The PI decision at `4da7564` fixes the target to the complete $O(\hbar^0)$ light–light phase in a stated scheme. This note tests the proposed integrability condition; it does not replace that target with an exact S-matrix programme.

Provisional choices (d) . Use the first higher residue, H_7, modulo lower charges, Wick products in the free good vacuum at $s = 1$, and the sharp, background-independent momentum cutoff of `good_renorm.tex`. More precisely, cut off vacuum contractions in local composite products, retaining the continuum canonical bracket and symmetric coincident-point products. This specifies a local subtraction/Ward prescription, not a finite-mode Hamiltonian truncation. This choice and the extra static matching recommendation are in README's Open decisions; neither is recorded as a new PI decision.

Order counting (c) (`integ_ordering.py`). We require the coefficient of $\hbar$ in $[\widehat{Q}, \widehat{H}]/(i\hbar)$ to vanish as well as its classical coefficient. The unreduced commutator then vanishes through $O(\hbar^2)$. For Weyl symbols and a constant canonical Poisson tensor, interchange of the symbols reverses odd contractions and preserves even ones. Hence

$$\frac{[\mathrm{Op}_W(F), \mathrm{Op}_W(G)]}{i\hbar} = \mathrm{Op}_W(\{F, G\}) + O(\hbar^2). \tag{1}$$

The script checks the second-order star-product cancellation explicitly. The third-contraction Moyal term would occur at $O(\hbar^2)$ in this *reduced* expression. It is beyond the requested order and is not computed here.

2 First higher charge in canonical variables

Canonical bracket (c) (`integ_charges.py`). Set $s = c^2$, $u = \phi_x$ and retain $\Omega = \int \mathrm{d}\pi \wedge \mathrm{d}\phi$, $\{\phi(x), \pi(y)\} = \delta(x-y)$. For local functionals of u, π,

$$H_s = \int h_s \mathrm{d}x, \qquad h_s = \tfrac{1}{2}\pi^2 + \tfrac{s}{2}u^2 - u^3 + \tfrac{1}{2}u_x^2, \tag{2}$$

$$\{F, G\} = \int \left[\frac{\delta F}{\delta u}\partial_x \frac{\delta G}{\delta \pi} + \frac{\delta F}{\delta \pi}\partial_x \frac{\delta G}{\delta u}\right] \mathrm{d}x, \qquad \frac{\delta F}{\delta \phi} = -\partial_x \frac{\delta F}{\delta u}, \tag{3}$$

$$u_t = \pi_x, \qquad \pi_t = (su - 3u^2 - u_{xx})_x. \tag{4}$$

Use a periodic line or decaying fields/variations with fixed potential endpoints. Total derivatives are discarded only under those conditions. The lower charges are $M = \int u\mathrm{d}x$, $\Pi = \int \pi \mathrm{d}x$, $P = -\int \pi u \mathrm{d}x$ and H_s; their conservation is checked separately. In the local u, π bracket M, Π are Casimirs; in potential variables Π generates the constant shift of ϕ.

Explicit representative (c) (`integ_charges.py`).

$$\boxed{Q_s = \int q_s \mathrm{d}x, \qquad q_s = \pi_x^2 + u_{xx}^2 - 3u\pi^2 + 3u^4 - 9uu_x^2 + s(u_x^2 - u^3).} \tag{5}$$

The π_x^2, u_{xx}^2 terms establish independence from the lower charges. Its variational derivatives are

$$\frac{\delta Q_s}{\delta \pi} = -6u\pi - 2\pi_{xx}, \tag{6}$$

$$\frac{\delta Q_s}{\delta u} = -3\pi^2 + 12u^3 + 9u_x^2 + 18uu_{xx} + 2u_{xxxx} - s(3u^2 + 2u_{xx}). \tag{7}$$

Substituting into the bracket gives a density whose Euler operators in both independent jet families vanish exactly, as does its constant part. It is therefore a total derivative:

$$\boxed{\{Q_s, H_s\} = 0.} \tag{8}$$

This is an arbitrary-field identity, not a soliton-only test.

Hierarchy identification (c) (`integ_charges.py`). The scalar Lax dictionary, with coefficients to the left of D_x, is

$$L = D_x^3 + aD_x + b, \qquad a = \tfrac{3}{2}u - \tfrac{s}{4}, \qquad b = \tfrac{3}{4}u_x + \tfrac{\sqrt{3}}{4}\pi, \qquad \tau = \sqrt{3}\,t, \tag{9}$$

$$a_\tau = 2b_x - a_{xx}, \qquad b_\tau = b_{xx} - \tfrac{2}{3}a_{xxx} - \tfrac{2}{3}aa_x. \tag{10}$$

Both equations reduce to (4). Define vacuum-subtracted residues $\mathsf{H}_n = \int[\operatorname{Res} L^{n/3} - \operatorname{Res} L^{n/3}|_0]\mathrm{d}x$. Independent pseudo-differential multiplication and the Riccati recursion $f_{xx} + 3ff_x + f^3 + af + b = z^3$, $f = z + \sum_{n\geq 1} f_n z^{-n}$, check $\mathsf{H}_n = -n\int(f_n - f_n|_0)\mathrm{d}x$ through $n = 7$, modulo derivatives. They give

$$\mathsf{H}_1 = \tfrac{1}{2}M, \qquad \mathsf{H}_2 = \tfrac{\sqrt{3}}{6}\Pi, \qquad \mathsf{H}_3 = \mathsf{H}_6 = 0, \tag{11}$$

$$\mathsf{H}_4 = -\tfrac{\sqrt{3}}{6}P - \tfrac{\sqrt{3}s}{36}\Pi, \qquad \mathsf{H}_5 = \tfrac{5}{24}H_s - \tfrac{5s^2}{288}M, \tag{12}$$

$$\mathsf{H}_7 = -\tfrac{7}{144}(Q_s + sH_s) + \tfrac{7s^3}{2592}M. \tag{13}$$

Thus (5) is the first higher hierarchy charge beyond the specified four. No second-bracket quantum normalization has been imported.

(c) Independently, the pulse $u = k^2 \operatorname{sech}^2(k(x - vt)/2)/2$, $\pi = -vu$, $v^2 = s - k^2$ gives

$$Q_s^{\mathrm{sol}} = 4k^5(20k^2 - 21s)/105. \tag{14}$$

Exact integration after $y = \tanh(k(x - vt)/2)$ and three independent 60-digit quadratures agree to better than 10^{-55}.

3 Test of the elastic direction before quantum ordering

Background shift (c) (`integ_charges.py`). The proposed (d) hypothesis is confirmed as a classical local identity:

$$s = 1 - 6U, \tag{15}$$

$$h_1(U + u, \pi) = h_s(u, \pi) + (U - 3U^2)u + U^2/2 - U^3, \tag{16}$$

$$q_1(U + u, \pi) = q_s(u, \pi) - 6Uh_s(u, \pi) + (12U^3 - 3U^2)u + 3U^4 - U^3. \tag{17}$$

The Hamiltonian changes by a mass charge and a constant; the higher charge changes by lower charges and a constant. The Lax dictionary verifies the same hierarchy flow for every s; $s > 0$ selects the stable linear good vacuum. The shift relates different background sectors, not a dynamical change of mean within a fixed-endpoint sector.

Explicit tangent charge (c) (`integ_charges.py`). Let $\mathcal{I} = \int u^2 \mathrm{d}x$, $\mathcal{J} = \int (u_x^2 - u^3)\mathrm{d}x$. Then

$$H_{1+2\hbar\lambda} = H_1 + \hbar\lambda\mathcal{I}, \qquad Q_{1+2\hbar\lambda} = Q_1 + 2\hbar\lambda\mathcal{J}, \tag{18}$$

$$\{Q_1, \lambda\mathcal{I}\} + \{2\lambda\mathcal{J}, H_1\} = 0, \qquad \{\mathcal{J}, \mathcal{I}\} = 0. \tag{19}$$

Here the subscript on H_s, Q_s is the sound coefficient; the quantum correction coefficient requested in the task is $Q_\lambda^{(1)} = 2\lambda\mathcal{J}$. The family is conserved exactly classically. In contrast $\{\mathcal{I}, H_s\} = 2\int u\pi_x \mathrm{d}x$ is generally nonzero: being an integrable direction does not make $\mathcal{I}$ a conserved charge.

A regulator-independent solvability observation. (c) (`integ_charges.py`, `integ_ordering.py`) Let $\mathcal{A}$ be a fixed one-loop local source at $\lambda = 0$. If R solves $\{R, H_1\} = -\mathcal{A}$, then

$$\{R + 2\lambda\mathcal{J}, H_1\} + \{Q_1, \lambda\mathcal{I}\} + \mathcal{A} = 0 \quad \text{for every } \lambda. \tag{20}$$

Contracting a counterterm already multiplied by $\hbar$ cannot change $\mathcal{A}$ at this order. Thus solvability is independent of λ: either all values admit a local correction, or an independent obstruction remains for all. Forbidding the tangent correction would impose an extra normalization on Q, not integrability. This argument is about a consistent continuum local Ward problem, not an arbitrary finite lattice.

4 All local one-loop ordering terms in this representative

Vacuum contractions (c) (`integ_ordering.py`). Use $\omega(q) = |q|\sqrt{1 + q^2}$ and $|q| \le \Lambda$ in contractions. With $R_\Lambda = \sqrt{1 + \Lambda^2}$, divide covariances by $\hbar$:

$$A = \langle u^2 \rangle_0/\hbar = (R_\Lambda - 1)/(2\pi), \tag{21}$$

$$B = \langle u_x^2 \rangle_0/\hbar = (R_\Lambda^3/3 - R_\Lambda + 2/3)/(2\pi), \tag{22}$$

$$C = \langle u_{xx}^2 \rangle_0/\hbar = (R_\Lambda^5/5 - 2R_\Lambda^3/3 + R_\Lambda - 8/15)/(2\pi), \tag{23}$$

$$D = \langle \pi^2 \rangle_0/\hbar = A + B = (R_\Lambda^3 - 1)/(6\pi), \tag{24}$$

$$E = \langle \pi_x^2 \rangle_0/\hbar = B + C. \tag{25}$$

Here D, E denote covariances. These are half-line integrals of $q/\sqrt{1+q^2}$, $q^3/\sqrt{1+q^2}$, $q^5/\sqrt{1+q^2}$, $q\sqrt{1+q^2}$, $q^3\sqrt{1+q^2}$, respectively, divided by 2π. All primitives and three cutoffs are checked by 60-digit quadrature. Symmetrized mixed u–π covariances and odd spatial coincidences vanish. The contraction matrix retains $\langle uu_{xx}\rangle_0/\hbar = -B$, although it contributes nothing for the chosen densities.

Wick-to-Weyl dictionary (c) (`integ_ordering.py`). For a polynomial f in the field jets set $Tf = \frac{1}{2}\sum C_{ab}\partial_a\partial_b f$. The Wick operator has Weyl symbol $f - \hbar Tf + O(\hbar^2)$. With $\mathcal{V} = \int \mathrm{d}x$ denoting only a vacuum constant,

$$\sigma_W(:H_s:) = H_s + 3\hbar AM - \tfrac{\hbar}{2}(D + sA + B)\mathcal{V} + O(\hbar^2), \tag{26}$$

$$\sigma_W(:Q_s:) = Q_s - 18\hbar A\mathcal{I} + \hbar(3D + 9B + 3sA)M - \hbar(E + C + sB)\mathcal{V} + O(\hbar^2). \tag{27}$$

In particular $-:u^3:= -u^3 + 3\hbar Au$ reproduces the Wick term in `good_renorm.tex`. At $s = 1$ the Hamiltonian constant is $-\hbar D\mathcal{V}$, the subtracted free zero-point energy. All nonconstant terms in Q_s arise from $-3u\pi^2$, $3u^4$, $-9uu_x^2$ and $-su^3$. No derivative or momentum operator is added at this order. Double self-contractions of u^4 start at $\hbar^2$.

The uncorrected charge fails (c) (`integ_ordering.py`). Mass and constant terms commute locally. Simple Wick ordering therefore leaves

$$\frac{[:Q_1:,:H_1:]}{i\hbar} = -18\hbar A\{\mathcal{I}, H_1\} + O(\hbar^2) = 36\hbar A\int \pi u_x \mathrm{d}x + O(\hbar^2). \tag{28}$$

A direct Wick double-contraction check gives the same source. The surviving nonlinear pair is $-3u(x)\pi(x)^2$ against $-u(y)^3$. With $r = x - y$, its second-contraction commutator divided by $i\hbar$ is $36\hbar C_{uu,\Lambda}(r)\delta'(r)\pi(x)u(y)$. Evenness gives $C_{uu,\Lambda}(r)\delta'(r) = A\delta'(r)$, which integrates to (28). Other double contractions vanish or are total derivatives. This prevents mistakenly dropping the source merely because the reduced *Weyl* commutator has no first-order Moyal term.

Local charge mixing (c) (`integ_ordering.py`). Choose

$$\widehat{H}_\lambda =: H_1 : +\hbar\lambda :\mathcal{I}:, \tag{29}$$

$$\widehat{Q}_\lambda = \boxed{:Q_1: +\hbar(18A :\mathcal{I}: +2\lambda :\mathcal{J}:)} + O(\hbar^2). \tag{30}$$

Equation (19) cancels the elastic deformation; $18A:\mathcal{I}:$ cancels (28). The remaining Weyl symbols are

$$\sigma_W(\widehat{H}_\lambda) = H_1 + \hbar[\lambda\mathcal{I} + 3AM - D\mathcal{V}] + O(\hbar^2), \tag{31}$$

$$\sigma_W(\widehat{Q}_\lambda) = Q_1 + \hbar[2\lambda\mathcal{J} + (6A + 12B)M - 2(B + C)\mathcal{V}] + O(\hbar^2), \tag{32}$$

$$[\widehat{Q}_\lambda, \widehat{H}_\lambda]/(i\hbar) = O(\hbar^2) \quad \text{in the stated local Ward prescription.} \tag{33}$$

Term	Interpretation	Consequence
Constants and M	Vacuum and lower-charge origins	Commute; choose consistently with the reference pressure.
$\hbar\lambda\mathcal{I}$ in H	Integrable elastic direction	Requires $2\hbar\lambda\mathcal{J}$ in Q; leaves λ free.
$-18\hbar A\mathcal{I}$ in $\sigma_W(:Q:)$	Nonconserved Wick mixing	Cannot be removed by conserved lower charges alone; needs the opposite local composite-charge term.
Conserved finite additions to Q	Hierarchy normalization	Allowed, but do not determine the elastic Hamiltonian coupling.

(c) The charge-mixing coefficient is fixed within this prescription, including its finite part: $18A = 9(\sqrt{1+\Lambda^2} - 1)/\pi$ and $18A - 9\Lambda/\pi \longrightarrow -9/\pi$. Removing only the leading divergence leaves a finite Ward source. There is no nonremovable *local first-order* obstruction here and no condition on λ.

Operator and regulator limitation (d) . The mixing diverges with the cutoff. Its cancellation in a local Ward identity does not establish convergence of interacting charge matrix elements, a positive representation, or boundary-compatible domains. In particular it does not prove existence of $\lim_{\Lambda\to\infty}\widehat{Q}_\lambda$ on the free Fock domain. A literal spectral projection also changes the canonical bracket and may break the classical hierarchy; its contact and cutoff-edge terms must be checked separately. Such an operator construction remains open. This does not restore a λ selection in the local problem: (20) makes the elastic tangent a homogeneous solution regardless of the inhomogeneous ordering source.

5 What fixes the remaining finite coupling?

Vacuum conditions (c) (`integ_observables.py`). With the pressure and energy origins of `good_renorm.tex`,

$$e(U) = \tfrac{1}{2}U^2 - U^3 + \hbar[W(U) + \lambda U^2] + O(\hbar^2), \tag{34}$$

$$W(U) = -\frac{(1-6U)^{3/2} - 1 + 9U}{6\pi}, \qquad U < 1/6, \tag{35}$$

$$e(0) = e'(0) = 0, \qquad K(0) = e''(0) = 1 + \hbar(2\lambda - 9/(2\pi)) + O(\hbar^2), \tag{36}$$

$$c_R := \sqrt{K(0)} = 1 + \hbar s_1 + O(\hbar^2), \qquad s_1 = \lambda - 9/(4\pi). \tag{37}$$

The curvature is checked by a convergent 70-digit integral and symbolic differentiation. Thus $K(0) = 1$, equivalently unit *static/hydrodynamic* speed to this order, fixes $\lambda = 9/(4\pi)$. A specified shift s_* would give $\lambda = s_* + 9/(4\pi)$.

(c) Fixed mean $U = 0$ specifies a sector; zero pressure there holds for every λ, because $(\lambda U^2)'|_0 = 0$. Neither fixes curvature. If instead $U = \hbar\rho$, the speed coefficient is $\lambda - 3\rho - 9/(4\pi)$; the reference mean must be stated.

(d) "Physical sound speed equals one" needs a definition here. The existing vacuum note distinguishes the thermodynamic speed from the sequential bare-on-shell slope and leaves the strict fixed-$\hbar$ spectral limit open. We do not identify these limits or repeat that spectral calculation. Static matching is an extra reproducible scheme choice; an exact pole/edge condition needs additional infrared input. Integrability and mean/pressure alone give no finite number for λ.

6 Sound units and invariant combinations

Classical scaling (c) (`integ_observables.py`). For $s = c^2 > 0$,

$$x = X/c, \quad t = T/c^2, \quad \phi = c\Phi, \quad u = c^2\Phi_X, \quad \pi = c^3\Phi_T \tag{38}$$

maps the action to c^3 times the unit-sound action. Thus

$$S_c = c^3 S_0, \quad \Omega_c = c^3\Omega_0, \quad P_c = c^4 P_0, \quad E_c = c^5 E_0, \quad \hbar_R = \hbar/c^3. \tag{39}$$

All transformed action coefficients, the symplectic factor and soliton energy/momentum are checked independently. This is a unit/parameter conversion, not a canonical transformation keeping numerical $\hbar$ fixed.

One body (c) , conditional on the audited trace (`integ_observables.py`). On the light branch take the existing conditional inputs

$$P_0(k) = \epsilon\tfrac{2}{3}k^3\sqrt{1-k^2}, \qquad E_0(k) = \tfrac{2}{3}k^3 - \tfrac{8}{15}k^5, \tag{40}$$

$$F_\lambda(k) = F_0(k) + \tfrac{2\lambda}{3}k^3, \tag{41}$$

$$F_0(k) = \frac{\sqrt{3}}{\pi}(\tfrac{3}{4} - k^2)\operatorname{artanh}\frac{2k}{\sqrt{3}} - \frac{3k}{2\pi}. \tag{42}$$

These are not a new determinant evaluation. At fixed momentum $E_0'(P) = v$ and $5E_0 - 4Pv = 2k^3/3$. With energy in c_R^5 units and momentum in c_R^4 units,

$$F_R = F_\lambda - s_1(5E_0 - 4PE_0') = F_0 + \frac{9}{4\pi}\frac{2k^3}{3}, \tag{43}$$

$$C_g = \lim_{k\to 0} F_\lambda/|P_0| = \lambda - 2/\pi, \qquad C_R = C_g - s_1 = 1/(4\pi). \tag{44}$$

In F_R, k labels the unit-sound solution at the rescaled momentum. The entire displayed F_R, not just its slope, is independent of λ within this elastic family.

Two bodies (c) , conditional on the stated frame (`integ_observables.py`). Use the chamber-reversing Darboux frame of `good_twobody.tex`, in which $\partial_{P_j}\delta_0 = -\Delta X_j'$. Rescaling gives

$$\delta_c(P_1, P_2) = c^3\delta_0(P_1/c^4, P_2/c^4), \tag{45}$$

$$K_\lambda = \partial_c\delta_c|_1 = 3\delta_0 - 4\sum_j P_j\partial_{P_j}\delta_0 = 3\delta_0 + 4\sum_j P_j\Delta X_j'. \tag{46}$$

Since $c = 1 + \hbar\lambda + O(\hbar^2)$ for the bare elastic deformation, this is the fixed-laboratory-momentum response per unit λ. The script differentiates the explicit root-cross-ratio phase and independently inverts deformed momenta before taking central differences. At $(\alpha_1, \alpha_2) = (0.3, 0.6)$ it finds $K_\lambda^{++} = 0.05008140486841890958\ldots$ and $K_\lambda^{+-} = -0.22517393868334175848\ldots$. All four orientations and exchange antisymmetry pass.

(c) Write the dimensionless phase as

$$\Theta(P;\hbar) = \delta_0(P)/\hbar + \delta_{1,\lambda}(P) + O(\hbar), \qquad \delta_{1,\lambda} = \delta_{1,0} + \lambda K_\lambda. \tag{47}$$

This specifies only the response; $\delta_{1,0}$ remains unknown. With $p_j = P_j/c_R^4$, $\hbar_R = \hbar/c_R^3$,

$$\Theta = \delta_0(p)/\hbar_R + \delta_{1,R}(p) + O(\hbar_R), \tag{48}$$

$$\boxed{\delta_{1,R} = \delta_{1,\lambda} - s_1 K_\lambda = \delta_{1,0} + \frac{9}{4\pi}K_\lambda.} \tag{49}$$

Changing momentum arguments alone misses the $3\delta_0$ term from action/$\hbar$ normalization and does not remove the entire response.

(c) These invariants are precisely independent along the specified elastic change, with kinetic/cubic/dispersive normalization, pressure origin and boundary/frame prescription held fixed. At fixed laboratory momentum, λK_λ is generally nonzero: changing λ without matching measured stiffness changes the model. This does not classify invariants under all finite counterterms or all quantum completions. **(d)** Boundary/state normalization and the independent two-body seed remain unfixed after sound-unit conversion.

7 Consequence for the target and proposed report changes

(d) Compute the requested phase provisionally in the *static reference-vacuum scheme*: free-good-vacuum Wick subtraction, $e(0) = e'(0) = 0$, $K(0) = 1$, hence $\lambda = 9/(4\pi)$, with one physical momentum and boundary prescription throughout. Retain the arbitrary-λ response for conversion. This is an additional normalization proposal, not an integrability prediction or an already accepted PI decision. Until accepted, λ remains free. Even after it is fixed, the absolute two-body seed is an unknown calculation, not a scheme choice.

The following is proposed text for the report's "Good-equation priorities" in Conclusions; `report.tex` is not edited.

1. **(c) /(d) Complete the scheme specification.** Record the higher-charge test's unrestricted λ, the explicit local correction and operator qualification. Adopt an extra vacuum condition; static $K(0) = 1$ is recommended. An exact spectral condition needs its infrared definition.

2. **(d) Calculate the complete light–light phase in that scheme.** Construct fixed-momentum physical states and evaluate the boundary-compatible, time-dependent projected determinant and measure. Retain the audited endpoint subtraction; check both λK_λ and (49). Factorisation/clustering alone does not determine the absolute seed.

3. **(d) Validate quantum charges and scattering channels.** Specify composite-charge matrix elements/domains and light-sector closure, including phonons and heavy resonances. Do not promote a local first-order identity to operator existence or all-loop integrability. Higher Moyal orders are outside this task.

4. **(d) Complete quantum defect theory as supporting work.** Determine boundary terms, normalization, absorption locus and residues in the same physical scheme. The existing deficient-rank tests cannot fix the bulk elastic coupling.

5. **(d) Keep exact candidate/ring work as a longer-term programme.** Physical poles, branches and quantum seams can follow a controlled one-loop phase; an exact S-matrix is not a prerequisite for the PI target.

8 Summary of claims and checks

Claim and status	Check and scope	Script
(c) First higher canonical charge	Exact Euler bracket; independent Riccati and pseudo-differential residues through H_7.	`integ_charges.py`
(c) Arbitrary sound coefficient	Both Lax equations, exact background shift, tangent identity and independent pulse integral.	`integ_charges.py`
(c) Local ordering terms	Full jet contraction matrix, exact primitives and 60-digit quadrature at three cutoffs.	`integ_ordering.py`
(c) Wick defect and correction	Nonzero uncorrected source, direct double contraction and corrected symbolic-λ Ward identity.	`integ_ordering.py`
(c) Static matching vs pressure	Energy derivatives, independent convergent curvature integral and shifted reference mean.	`integ_observables.py`
(c) Conditional sound invariants	Action/symplectic/momentum factors, full one-body elastic subtraction and inverse-momentum phase response.	`integ_observables.py`
(d) Quantum operator and phase	No assertion of charge convergence, positive representation, exact spectral speed or completed absolute phase.	Explicit open scope

9 Failures recorded and execution record

1. **(c) Naive Wick ordering fails.** Equation (28) is an explicitly checked nonzero source, retained as a negative test. Adding only the λ tangent correction does not cure it. The stated composite mixing does; no check was weakened.

2. **Two development assertions used structural equality.** The first `integ_charges.py` run compared an expanded Euler derivative to an unexpanded polynomial; the first `integ_observables.py` run compared algebraically equal forms of the small-k limit. Both failed and were repaired by testing that the expanded/simplified difference is exactly zero. Mathematical claims and numerical tolerances were unchanged.

3. **Repository infrastructure.** The initial shared-checkout pull failed because `.git/FETCH_HEAD` is read-only. Work used the writable `/tmp/integ_scheme_work` clone on `main`, with the project remote and a successful initial pull. An initial document-write command and a later script-edit command used duplicated relative directories; neither wrote its target, and both were rerun with correct paths. The first PDF build also caught a box spanning an alignment separator; the formula layout was corrected. An early execution-record assembly was rejected by its suite-completion guard; the record is assembled only after all scripts finish. Only task files are committed; historical regenerated outputs remain in scratch.

Fresh checks. All three new scripts end with `ALL CHECKS PASSED`. The largest observed errors are 6.08×10^{-64} for the charge quadrature, 3.89×10^{-62} for the vacuum contractions, and 1.62×10^{-35} for the fixed-momentum phase response. Their asserted bounds are 10^{-55}, 10^{-53} and 10^{-32}, respectively.

Full historical regression. All 129 top-level historical Python scripts were run sequentially from their own directories in `phase1/scripts`, `phase2/scripts` and `corrigan-zambon`. Of these, 127 returned exit code zero and 2 failed. 2 zero-exit scripts printed no success marker; they are not reclassified as passing verification merely from their exit code. The per-script exits, runtimes, marker observations and failed log tails are in `scripts/integ_regression_results.json`, with an inventory entry in `../scripts_inventory.md`. Nested script runs are not counted again. All historical source files and tolerances are unchanged; regenerated caches/figures stay in scratch. Failures are:

- `phase2/scripts/audit_onebody.py`: exit 1. The existing line-383 extrapolated-identity bound 3×10^{-6} fails, as already documented in README and the inventory.
- `phase2/scripts/toda_edge_twoterm.py`: exit 1. The existing successive-raw-value bound 0.004 fails, already documented in the inventory; it was not relaxed.

10 Sources and reading record

No external model-specific theorem or bibliographic identifier is used. Results are project derivations (c), with conditional inputs labelled. The following repository texts and relevant scripts were read directly:

- `AGENTS.md`, `README.md`, PI decision `4da7564`.
- `phase1/review.tex`, Lax/hierarchy sections; its dictionary is rederived above, not used to assume canonical quantum ordering.
- `phase2/good_renorm.tex`, Wick subtraction, static elasticity, spectral-limit qualification and conditional trace.
- `phase2/good_twobody.tex`, counterterm/kernel scripts, and `phase2/report.tex`, fixed-momentum scattering and Conclusions.
- `phase2/shiftcharge_quantum_bridge.tex`, `shiftcharge_bridge_hierarchy.py` and `resonant_channels_conservation.py`, for existing charge conventions.

New checkers import no historical checker; only `integ_ordering.py` imports the new charge script's jet-algebra utilities. No second-bracket quantum result is transferred to the canonical theory.